

Rational design of hierarchical zeolite encapsulating FeMnK architecture to enhance light olefins selectivity in Fischer-Tropsch synthesis

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ABSTRACT

Construction of core-shell catalysts is generally considered as an effective strategy for regulating hydrocarbon distribution in Fischer-Tropsch synthesis. Rational design of core-shell catalysts to enhance olefins selectivity is greatly desirable but still challenging. Herein, a series of FeMnK@H-S-1(x) catalysts (FeMnK oxide encapsulated in hollow silicalite-1 zeolite) featured with hierarchical shell were elaborately fabricated by “desilication-recrystallization” method in a mixed alkaline solution ($x = \text{NaOH}/(\text{TPAOH} + \text{NaOH})$). Hierarchy factor was introduced to quantitatively describe variety of hierarchical zeolite encapsulating structures. In catalytic performance, it was found that the light olefins selectivity attained the maximum at 50.6% over FeMnK@H-S-1(0.20) catalyst with space-time yield of light olefins up to $26.7 \mu\text{mol}_{\text{C}_2=\text{C}_4} \cdot \text{g}_{\text{Fe}}^{-1} \cdot \text{s}^{-1}$ and performed a monotonicity increase relation with the hierarchy factor. Maximizing the hierarchy factor was beneficial to the formation of light olefins on the as-synthesized catalysts, which may result from the well coordination of shape-selective function and transport efficiency due to the optimally orchestrated porosities. Additionally, the moderate H_2 activation capacity, relatively lower olefins desorption energy and the suppressed olefins secondary reaction over FeMnK@H-S-1(0.20) catalyst also contribute to the boosted light olefins selectivity. This study sheds new clues for rational catalyst design in direct light olefins synthesis from Fischer-Tropsch synthesis.

1. Introduction

Light olefins including ethylene, propylene and butene are important primary materials for the synthesis of various downstream products in the chemical industry, such as solvents, polymers and cosmetics. Traditionally, light olefins are mainly obtained from the catalytic or pyrolysis of crude oil. With the decrease of crude oil stock and the continuous increase of demand for light olefins, CO hydrogenation reaction provides a non-petroleum alternative route for the synthesis of light olefins from natural gas, coal, biomass and organic wastes [1]. In this process, syngas ($\text{CO} + \text{H}_2$) as an important intermediate of non-petroleum carbon resources, is generally used to produce light olefins through two different routes. One is the indirect route that syngas firstly converts into methanol or dimethyl ether and then produces olefins. The other is the direct conversion of syngas to light olefins, including OX-ZEO process which combines methanol synthesis and methanol-to-olefin (MTO) tandem reaction and Fischer-Tropsch synthesis to olefins (FTO) process [2,3].

In the past decade, the indirect synthesis of light olefins has been

commerized as the relatively lower technical threshold of methanol or dimethyl ether conversion. Nevertheless, the direct route via syngas is more appealing due to its short process and economic benefits [1]. Recently, the research on the process for direct light olefins synthesis has obtained some important progress. For example, researchers have developed novel OX-ZEO catalysts (e.g., bifunctional ZnCrO_x -SAPO-34 and ZrZn -SAPO-34 catalysts) to enable the above indirect processes to be efficiently carried out in one reactor to synthesize light olefins [4,5]. However, the uneven mixed catalysts and the strict reaction window coordinating the consecutive reactions of methanol synthesis and MTO processes seem to be a challenge for engineering design. Another direct olefins synthesis route is modifying the F-T synthesis reaction to maximize the light olefins selectivity, namely FTO process. So far, many metal catalysts concentrated in group VIII metals, especially Fe [6–9], Co [10–12] and Ru [13,14] based catalysts have been extensively investigated for tuning product distribution. Among them, the iron-based catalysts have great potential in FTO process due to their preferable light olefins selectivity, high availability and improved water-gas-shift (WGS) activity. Additionally, promoters are of great significance

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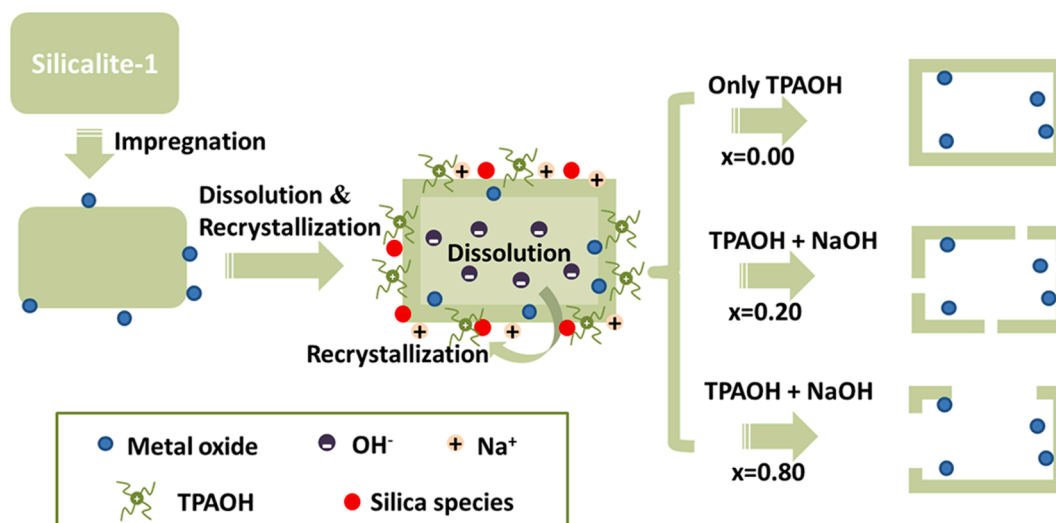
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Scheme 1. The synthesis procedure for FeMnK@H-S-1(x).

for enhancing the selectivity of light olefins on iron-based catalysts. In previous studies, Mn and K have been reported to be the most effective promoters for iron-based catalysts in F-T synthesis to light olefins process [2,15–18]. Even though a lot of efforts have been made, there are still many problems, like wide product distribution and unsatisfactory light olefin selectivity, needing to be further resolved.

Encapsulation of iron-based catalyst seems to be an effective strategy to regulate the wide product distribution resulting from uncontrollable carbon chain growth [19–21]. Meanwhile, microporous zeolites with well-defined channels have been often reported to be an excellent encapsulating shell candidate due to their shape-selective effect [22–24]. Nonetheless, the micropores of traditional zeolites usually depreciate the catalytic activity and aggravate the possibility of secondary hydrogenation of olefins in practice owe to reactants and products transfer limitation. To alleviate this drawback, encapsulating metal nanoparticles into hierarchical (micro-meso-macroporous) zeolite was freshly tried in FTO process [25], where the metal oxide acts as catalytic active site, the zeolite micropores domain the carbon distribution of olefins product, and the introduced macro-/mesopores ensured high diffusion efficiency. However, few studies have systematically depicted the characteristics of hierarchical zeolite encapsulating catalysts with the optimal balance between shape-selective effect and transport properties to match the requirements of light olefins synthesis.

Herein, a series of hierarchical zeolite encapsulating FeMnK catalysts involving micro-/meso-/macropores were rationally designed by “desilication-recrystallization” method through the adjustment (TPAOH + NaOH) mixed alkaline solution, as shown in Scheme 1. Hierarchy factor was introduced to quantitatively describe variety of as-synthesized hierarchical zeolite encapsulating structure and it was found that the catalytic activity and light olefins selectivity are closely related to the hierarchy factor. By characterizing the physicochemical properties of the catalyst, the effect of hierarchy factor, hydrogen activation capacity, olefins desorption energy and secondary reaction probability on catalytic performances is discussed and we gain insights into the key factors for light olefin synthesis in FTO process over hierarchical zeolite encapsulating catalysts.

2. Experimental section

2.1. Catalyst preparation

2.1.1. Synthesis of silicalite-1 and FeMnK/silicalite-1

The synthesis of silicalite-1 zeolite was referred to a previously reported method, using tetraethyl orthosilicate (TEOS) as the silicon

source and tetrapropylammonium hydroxide (TPAOH, 25%) as the template [26]. The synthesis gel with the elemental compositions of $\text{SiO}_2:0.27\text{TPAOH}:37\text{H}_2\text{O}$ was firstly stirred for 6 h at 35 °C and then placed in an autoclave, followed by crystallizing at 170 °C for 72 h. The as-synthesized silicalite-1 zeolite was gathered by centrifugation, washing several times with deionized water. After drying (80 °C, overnight) and calcination process (550 °C, 6 h), the final product was obtained.

The FeMnK/Silicalite-1 composites were prepared through an impregnation method. Typically, 1 g silicalite-1 zeolite was impregnated with 2 mL solutions of iron, manganese, and potassium nitrates ($\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $\text{Mn}(\text{NO}_3)_2$ (50 wt%) and KNO_3). Then, the water was evaporated at 80 °C with stirring. The residual solid product was dried overnight, and calcinated in static air at 400 °C for 4 h to get FeMnK/Silicalite-1 composite.

2.1.2. The synthesis of FeMnK@H-S-1(x)

Encapsulation of FeMnK oxides into hollow silicalite-1 zeolite with different pore sizes were synthesized by “desilication-recrystallization” method with mixed alkaline (TPAOH + NaOH) solution. Scheme 1 represents the synthesis procedure for FeMnK@H-S-1(x) catalysts, where x represents the molar ratio of NaOH in the mixed alkaline solution. First, dispersed 4.5 g FeMnK/Silicalite-1 composite into 30 mL mixed alkaline solution with varied NaOH/(TPAOH + NaOH) molar ratio while keeping the total OH^- concentration constant equal to 0.56 M. After stirring for 3 h at 25 °C, the mixture was put into an autoclave and heated to 170 °C for 24 h. Then, the brown solid product was recovered by centrifugation and washed with deionized water until natural pH. At last, the resultant product was dried overnight at 80 °C and then calcinated at 450 °C for 6 h.

2.2. Catalysts characterization

The texture properties and pore size distribution of FeMnK@H-S-1(x) catalysts were measured by a JW-BK200 analyzer at −196 °C. The XRD pattern was determined by the Bruker D8 diffractometer with Cu Kα ($\lambda = 1.54 \text{ \AA}$) radiation (40 kV, 40 mA). The morphology of as-investigated catalysts was obtained through transmission electron microscope (TEM, JEOL-2100) and scanning electron microscope (JSM-7001F). XPS characterization was conducted on the Thermo ESCALAB 250XI K-Alpha spectrometer. The element content of FeMnK@H-S-1(x) catalysts was derived from Thermo-Fischer XRF-3600. The catalysts (100 mg) were loaded in a U-type quartz reactor for TPR experiments in 5% H_2/N_2 flow at $30 \text{ mL} \cdot \text{min}^{-1}$. The hydrogen consumption patterns were detected by a

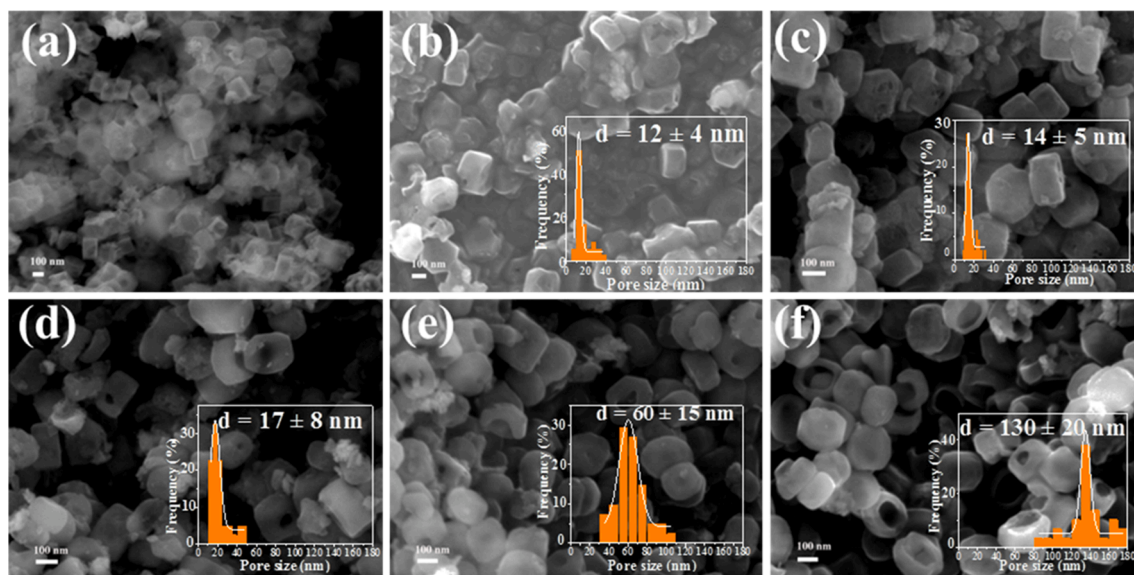


Fig. 1. SEM images of FeMnK@H-S-1(x) catalysts (inset: pore size distribution); (a) FeMnK@H-S-1(0.00), (b) FeMnK@H-S-1(0.10), (c) FeMnK@H-S-1(0.20), (d) FeMnK@H-S-1(0.50), (e) FeMnK@H-S-1(0.67), (f) FeMnK@H-S-1(0.80).

TCD, as increasing the temperature from 25 to 850 °C under 10 °C/min.

The $\text{H}_2/\text{C}_2\text{H}_4$ -TPD signals were monitored by Omni STAR 200 mass spectrometer. First, the samples underwent a reduction treatment under 20% H_2/N_2 flow at 400 °C. After cooling down in Ar flow, the catalyst was saturated with H_2 or C_2H_4 . Next, the system was purged with Ar to wipe off physically adsorbed molecules. The desorption curve of H_2 and C_2H_4 was recored under Ar flowing with increasing the temperature from 25 to 850 °C.

The pulse experiment was performed in the above mentioned quartz reactor, and monitored by Omni STAR 200 mass spectrometer. First,

spent catalyst (0.3 g) was purified in Ar flow at 280 °C to wipe off impurities remaining on the catalyst after the reaction. Next, the system was switched to hydrogen flow, and pulsed 0.8 mL of C_2H_4 every 10 min.

2.3. Catalytic evaluation

The performance of FeMnK@H-S-1(x) catalysts were evaluated in a fixed bed reactor at 280 °C, 0.5 MPa, 4000 h^{-1} , $\text{H}_2/\text{CO} = 1.9$. Usually, 1.5 mL catalyst (about 0.85 g) was loaded into the isothermal zone, and then reduced by 20% H_2 + 80% N_2 mixed gas under atmospheric pressure

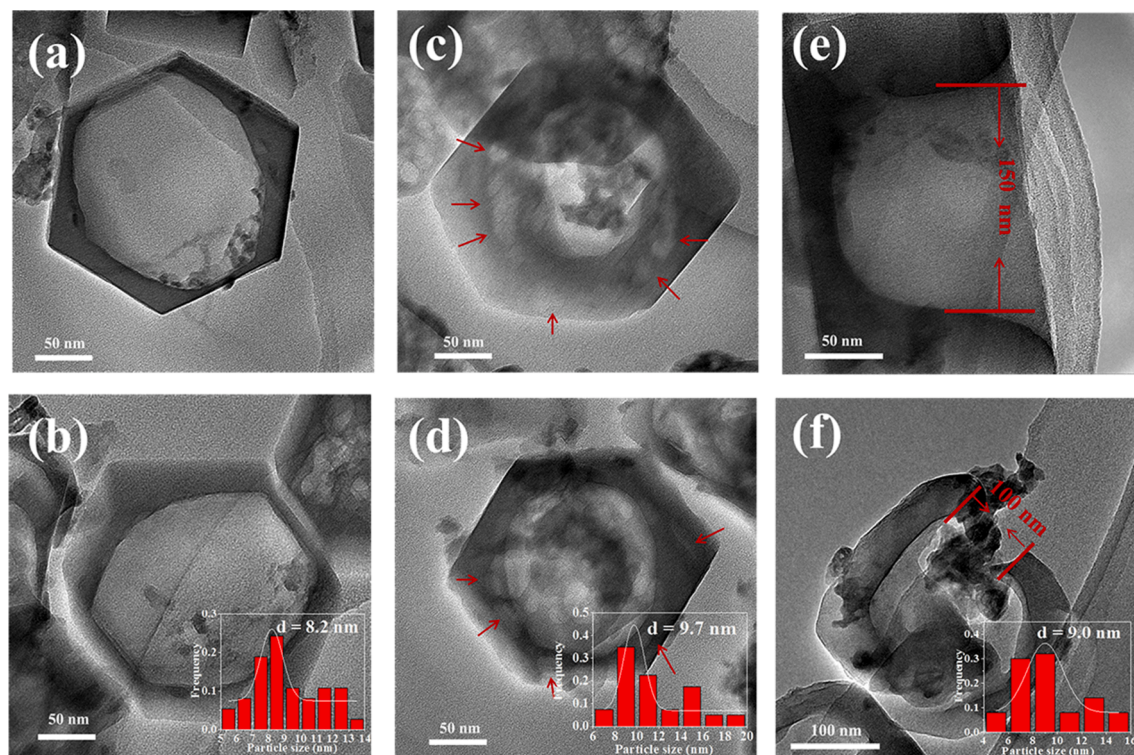


Fig. 2. TEM images of FeMnK@H-S-1(x) catalysts (inset: particle size distribution of metal oxide): (a and b) FeMnK@H-S-1(0.00), (c and d) FeMnK@H-S-1(0.20), (e and f) FeMnK@H-S-1(0.80).

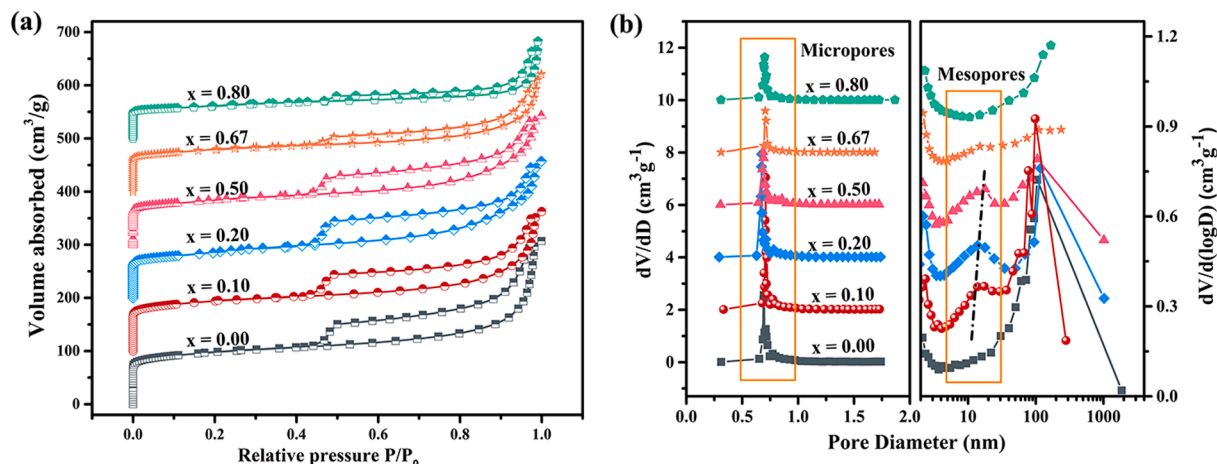


Fig. 3. (a) N_2 adsorption/desorption isotherms of different FeMnK@H-S-1(x) catalysts. All samples have been shifted upward by $100 \text{ cm}^3 \text{ g}^{-1}$ in sequence. (b) Micropore and mesopore size distribution of FeMnK@H-S-1(x) catalysts.

at 400°C for 6 h. After reduction, the syngas was switched and pressurized to the set value. Next, the reactor was raised to the specified temperature for reaction. The tail gas was analyzed by FID detector with Carbosieve packed column and TCD detector with Al_2O_3 column. Use FID detector with the DM-1 capillary column to analyze the collected oil.

3. Results and discussion

3.1. Synthesis and characterization

Fig. 1 presents the SEM images and corresponding pore size (mesopore and macropore) distribution of FeMnK@H-S-1(x) catalysts. It can be seen that there are no obvious meso-/macropores in the zeolite shell of FeMnK@H-S-1(0.00) catalyst with only TPAOH in the as-prepared alkaline solution (Fig. 1a). With the increase of NaOH fraction in the range of $0.10 \leq x \leq 0.50$ (x represents the molar ratio of $\text{NaOH}/(\text{TPAOH} + \text{NaOH})$), additional mesopores were formed in the crystalline silicalite-1 shell, and its size increased from 12 to 17 nm (see Fig. 1b–d). Further increasing NaOH proportion ($x \geq 0.67$), the mesopores in the crystalline silicalite-1 shell dramatically evolve into macropores (Fig. 1e and f). The FeMnK@H-S-1(0.67) and FeMnK@H-S-1(0.80) catalysts exhibit a large number of cup-shaped macropores with the sizes of ca. 60 nm and 130 nm, respectively. These suggested that a certain amount of NaOH in alkaline solution ($0.10 \leq x \leq 0.50$) could facilitate the generation of mesopores. Nevertheless, the excessive desilication would take place when NaOH concentration in the prepared alkaline solution increased to $x \geq 0.67$, and result in the severe destruction of zeolite shell, contributing to the formation of macropores. More importantly, the poly-type hierarchical pore constituted with micropore, mesopore/macropore in the silicalite-1 shell could be precisely regulated as desired by adjusting the ratio of NaOH in mixed alkaline solution.

TEM analysis was performed to confirm the hollow structure of the zeolite shell and provide detailed information about the location of metal nanoparticles in the FeMnK@H-S-1(x) catalysts. As shown in Fig. 2a and b, for FeMnK@H-S-1(0.00) catalyst, most FeMnK oxide particles are encapsulated by an intact hollow silicalite-1 shell with a thickness about 10–40 nm. Upon NaOH introduction in the as-prepared alkaline solution, a hollow structure with relative lower crystallinity and many bright voids in the shell (as marked by the red arrow) are observed over FeMnK@H-S-1(0.20) catalyst (Fig. 2c and d), demonstrating the existence of mesopores in the zeolite shell of catalyst. Furthermore, it could also be observed that almost all the metal oxide nanoparticles are located inside the mesoporous silicalite-1 shell with wall thickness about 10–40 nm. To further identify the encapsulation structure of the catalyst, FeMnK@H-S-1(0.10) catalyst was characterized by a high-angle

circular dark field scanning transmission electron microscope (HAADF-STEM), shown in Fig. S1. The HAADF-STEM image and the results of EDX elemental mapping (Fe, Mn, K, Si, O) show that the Fe, Mn metals are located at the inner wall of the hollow silicalite-1 shell, confirming the encapsulated structure. However, for FeMnK@H-S-1(0.80) catalyst (Fig. 2e and f), it can be seen that the macropores were created on the surface of zeolite shell, in line with SEM analysis. Accompanying result is that some of the nanoparticles are still situated at inner wall of the silicalite-1 shell, while some escape through the larger macropores. The metal oxide nanoparticles size distribution of FeMnK@H-S-1(0.00), FeMnK@H-S-1(0.20) and FeMnK@H-S-1(0.80) catalysts (inset of Fig. 2b, d and f) present a mean diameter of 8.2, 9.7 and 10.0 nm, respectively. It is obviously suggested that, despite their different structural characteristics, these catalysts still exhibit similar metal particle sizes. Additionally, they also have similar active metal contents confirmed by XRF (Table S1).

Fig. 3a shows the N_2 adsorption/desorption isotherms of FeMnK@H-S-1(x) catalysts. All the catalysts display an enhanced N_2 uptake at low relative pressure ($P/P_0 < 0.02$), signifying the presence of microporous structure with HK (Horvath-Kawazoe Fig. 3b, left) pore size distribution centered at 0.6–0.7 nm. Distinct H2-type hysteresis loops were observed on FeMnK@H-S-1(x) ($0.00 \leq x \leq 0.50$) catalysts with abrupt step at about $P/P_0 = 0.45$, corresponding to the capillary condensation of nitrogen in the large void with relative intact shells only accessible by smaller pores [27,28]. As the concentration of NaOH further increases ($x > 0.50$), the hysteresis loop slowly shrinks. This indicated that the size and amount of meso-/macroporous holes of the shell greatly increase, allowing the easy penetration of adsorption gases into the cavities.

According to the results of the BJH pore distribution plot (Fig. 3b, right), when only TPAOH is in the alkaline solution (FeMnK@H-S-1(0.00)), no pore size distribution in the mesoporous region (2–50 nm) is observed. When a small amount of NaOH is introduced (FeMnK@H-S-1(0.10)), a weak mesopore size distribution appears. Increasing the NaOH concentration to $x = 0.20$ and 0.50 , the mesopore size distribution becomes obvious, and the mean pore size of the latter is slightly larger than the former. However, when the concentration of NaOH further increases ($x \geq 0.67$), the mesopore size distribution becomes relatively wider, and finally disappears in the FeMnK@H-S-1(0.80) catalyst, confirming that excessive NaOH could destroy the mesoporous system, forming macropores. In short, the pore size of the zeolite shell depends on the composition of the alkaline solution used in the preparation of FeMnK@H-S-1(x). A suitable concentration of NaOH helps to form mesopores, while the larger one promotes the formation of macropores.

The relationship between the textural parameters of FeMnK@H-S-1(x) catalysts and the molar ratio of NaOH in the mixed alkaline

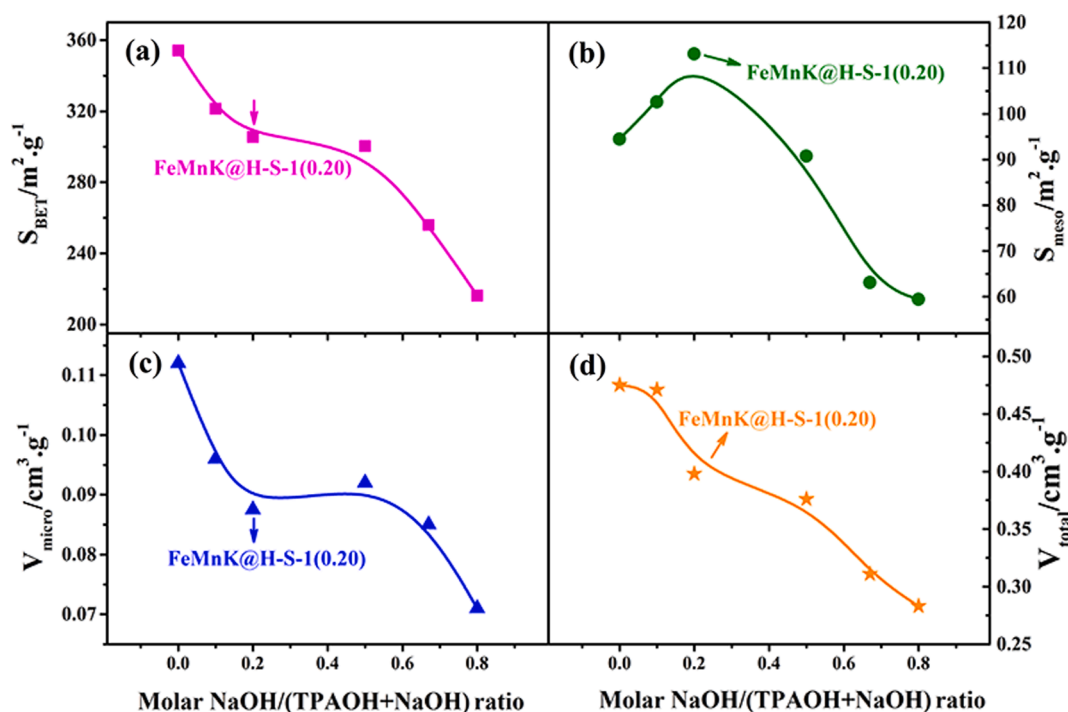


Fig. 4. Evolution of the textural parameters of FeMnK@H-S-1(x) catalysts in the function of NaOH/(TPAOH + NaOH) molar ratio: (a) total surface area, S_{BET} (b) mesopore surface area, S_{meso} (c) micropore volume, V_{micro} (d) total pore volume, V_{total} . S_{BET} derived from BET method. S_{meso} and V_{micro} measured by t-plot method. V_{total} taken from N_2 adsorbed at relative pressure of 0.99.

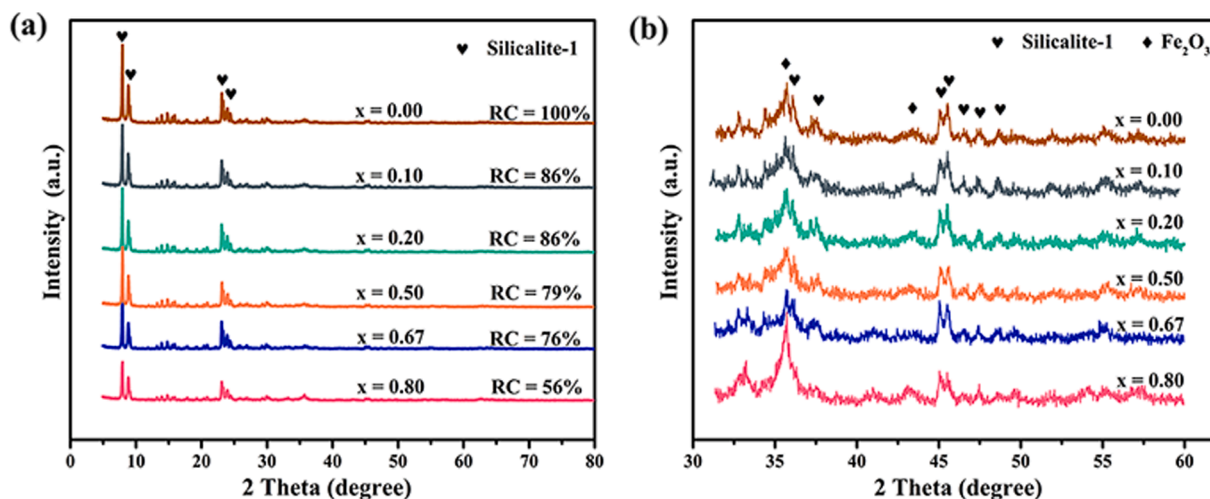


Fig. 5. (a) XRD patterns of FeMnK@H-S-1(x) catalysts; (b) Magnification of XRD patterns between 30 and 60°. Relative Crystallinity (RC) = $\frac{\text{Intensity of FeMnK@H-S-1}(x)}{\text{Intensity of FeMnK@H-S-1}(0.00)} \times 100\%$.

solution was plotted in Fig. 4 (the specific data listed in Table S2). The total surface area and total pore volume decrease linearly with the increase of NaOH fraction (Fig. 4a and d). As to the micropore volume (Fig. 4c), it declines first ($0.00 \leq x < 0.20$), then remains stable between $0.20 \leq x \leq 0.50$, and finally decreases significantly ($x > 0.50$). It should be noted that the mesopore surface area ($S_{\text{meso}} = S_{\text{BET}} - S_{\text{micro}}$) displays a volcanic dependency curve (Fig. 4b), with maximal values about $113 \text{ m}^2 \cdot \text{g}^{-1}$ in the FeMnK@H-S-1(0.20) catalyst, in which the mesopore surface area was enhanced and simultaneously the V_{micro} was largely preserved. At lower NaOH concentration ($x \leq 0.10$) the hollow microporous structure of the zeolite is not significantly changed, with V_{micro} , S_{BET} and V_{total} remaining at a higher level, but the mesoporous surface area is lower. On the other hand, at higher NaOH concentration ($x >$

0.50), all parameters obviously decrease, due to the formation of macropores caused by excessive desilication.

XRD patterns of FeMnK@H-S-1(x) catalysts are exhibited in Fig. 5. It should be noted that their typical MFI structures are well preserved after alkali treatment (Fig. 5a). However, the relative crystallinity decreases as the value of x increases, suggesting that the crystal structure of silicalite-1 zeolite is destroyed in different extent. The diffraction peaks between 30 and 60° (Fig. 5b) could be related to Fe_2O_3 crystal (JCPDS 25-1402), demonstrating the presence of Fe_2O_3 phase in the as-prepared catalysts [23,29]. But no peaks related to manganese are observed, which might be attributed to the well dispersed Mn species or the similar diffraction peaks to iron oxide.

XPS characterizations were conducted to analyze chemical states of

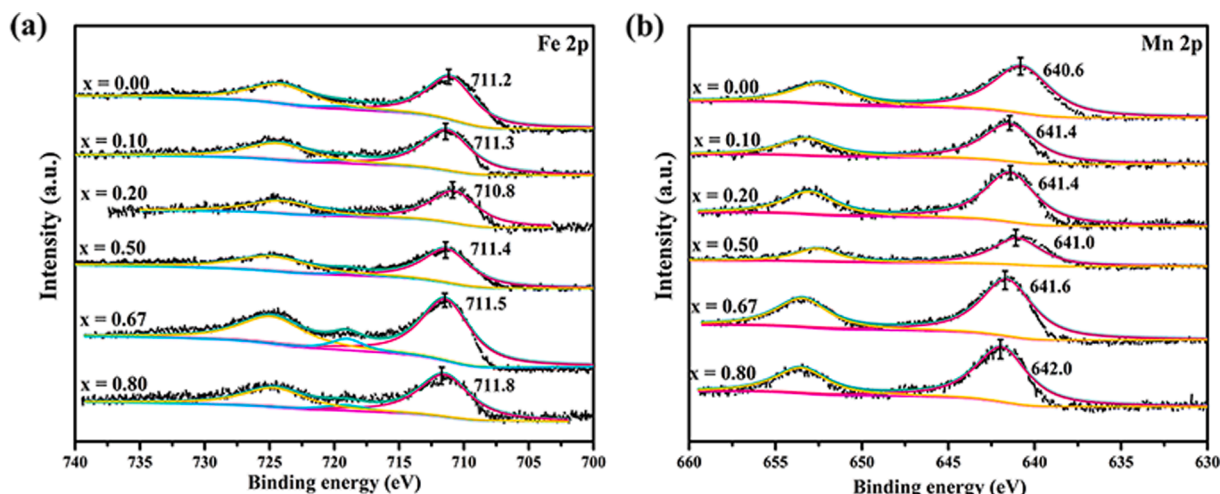


Fig. 6. (a) Fe 2p, (b) Mn 2p, XPS spectra of different FeMnK@H-S-1(x) catalysts.

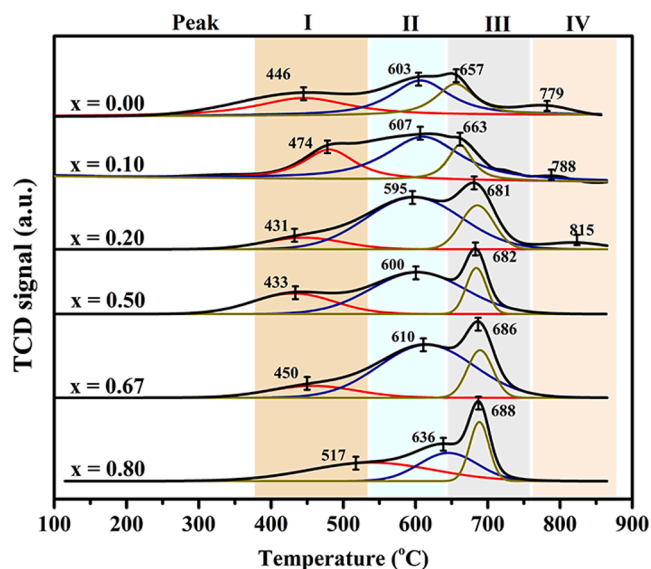


Fig. 7. H₂-TPR curves of the FeMnK@H-S-1(x) catalysts.

Fe, Mn and Si species over different FeMnK@H-S-1(x) catalysts (Figs. 6a, b and S2). As presented in Fig. 6a, the Fe 2p_{3/2} and 2p_{1/2} peaks of as-investigated catalysts all locate at near 711.0 and 724.4 eV, respectively, with a shakeup satellite peak at around 719.0 eV, which is corresponding to Fe₂O₃ phase [18,30]. The Mn 2p_{3/2} and 2p_{1/2} peaks locate at 640.6–642.0 and 652.2–653.6 eV, respectively, being attributed to the Mn₂O₃ phase [18,31]. The bind energy of Fe 2p_{3/2} of FeMnK@H-S-1 (0.00) catalyst (711.2 eV) is higher than that of bulk Fe₂O₃ (710.8 eV) [32,33], that of Mn 2p (640.6 eV) is lower than the value of bulk Mn₂O₃ (642.0 eV) [16,33], while that of Si 2p is similar to the standard value of silicalite-1 (103.3 eV) [34]. The above difference in binding energy could be ascribed to the electron transfer from Fe to Mn atoms. Huang et al. [18] also observed the similar electron transfer phenomenon in the FeMnCu/MCF-27 catalyst. With the increase of the mesopores size, the binding energy of Fe 2p decreases to 710.8 eV over FeMnK@H-S-1(0.20) catalyst, while that of Mn 2p increases to 641.4 eV, which are close to the value of bulk Fe₂O₃ and Mn₂O₃, indicating that the interaction between Fe and Mn species was weakened. However, further increasing the shell pore size (0.5 < x ≤ 0.8), the binding energy of Fe 2p moves upward. Meanwhile, the binding energy of Si 2p shifts downwards (see Fig. S2) and that of Mn 2p is close to the standard value of bulk Mn₂O₃.

This result indicates that a strong interaction between iron and silicalite-1 was constructed, due to the decrease of crystallinity of zeolite, as evidenced by XRD characterization (Fig. 5). Previous reports also observed that the interaction between metal–silica is higher than that of metal–zeolite [35]. The reason for this possibly is that more stable Si–O bonds exist in crystalline framework and resist formation of strong metal–support interactions. As the XPS characterization is only sensitive to the surface with a detection depth of a few nanometers [36], the concentration of surface Fe species could be obtained from the XPS spectrum, and listed in Table S1. Only about 1.0 wt% of Fe species were detected on the FeMnK@H-S-1(x) (x < 0.50) catalysts with relative small pores, far below its actual content (8.0 wt%) judging from XRF, which further confirmed that most of the metal oxides are encapsulated by the hollow zeolite shell.

Hydrogen-temperature programmed reduction (H₂-TPR) experiments were performed to investigate the reduction behavior of FeMnK@H-S-1(x) catalysts. All catalysts mainly include three obvious reduction peaks (I, II and III), and a weaker peak IV of some catalysts (see Fig. 7). It was reported that the progressive reduction of the iron oxide occurs in the sequence of (Fe, Mn)₂O₃ → (Fe, Mn)₃O₄ → (Fe, Mn)O → metallic Fe and MnO [37,38]. The temperature zone of 400–550 °C (peak I) can be ascribed to the reduction of (Fe, Mn)₂O₃ to (Fe, Mn)₃O₄. The distinct peaks II and III locating at 550–750 °C could be assigned to the reduction process of (Fe, Mn)₃O₄ to (Fe, Mn)O and then to metallic Fe and MnO. The additional reduction peak IV of FeMnK@H-S-1(x) (x ≤ 0.20) catalysts might be corresponding to the reduction of internal metal oxides or partially FeO strongly interacting with zeolite. By comparing the reduction peaks of FeMnK@H-S-1(x) catalysts, we find that pore size has an important influence on the reduction properties of the catalyst. Compared with FeMnK@H-S-1(0.00) and FeMnK@H-S-1(0.10) catalysts, the reduction peaks I and II of FeMnK@H-S-1(0.20) shifted to lower temperature, indicating that appropriate shell pores size can promote the reduction of metal oxides. With continuously increasing the pores size from mesopore to macropore (increasing the x value from 0.50 to 0.80), the reduction peaks gradually move to higher temperature, which is in accordance to the XPS results, demonstrating that the decrease in the crystallinity of the zeolite shell could enhance the interaction between metal–support.

3.2. Catalytic performance

Before discussing the performance of FeMnK@H-S-1(x) catalysts, we first investigated the reference supported FeMnK/S-1 catalyst. As shown in Table S3, the FeMnK/S-1 catalyst presents a CO conversion of 15.2% and hydrocarbon selectivity for CH₄, C₂[−]–C₄[−], C₂⁰–C₄⁰ and C₅⁺ of 15.8%,

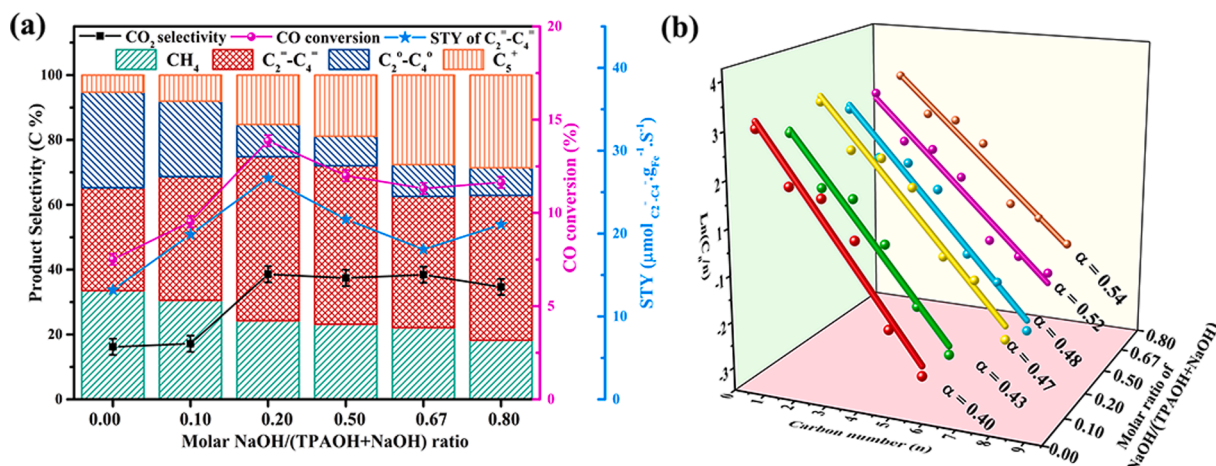


Fig. 8. (a) Product distribution, CO conversion and space-time-yield (STY) for C₂-C₄; (b) ASF plots of FeMnK@H-S-1(x) catalysts.

33.9%, 6.5%, and 45.8%, respectively. The high selectivity to C₅⁺ hydrocarbons is in accordance with the previously reported on this type of catalysts [39]. The performance of FeMnK and FeMnK + H-S-1 (physical mixing FeMnK oxide and hollow silicalite-1 zeolite) catalysts were also evaluated and listed in Table S4. It can be seen that they exhibit similar catalytic performance to FeMnK/S-1. As reported by our group, the use of encapsulated structure catalysts could significantly reduce the selectivity of C₅⁺ hydrocarbons and promote that to the CH₄ and C₂-C₄ hydrocarbons [25]. In present work, we find that the product selectivity is closely related to the composition of alkaline solution used for FeMnK@H-S-1(x) catalysts, as presented in Fig. 8a. For the FeMnK@H-S-1(0.00) and FeMnK@H-S-1(0.10) catalysts prepared using TPAOH or mixed alkali solution with high TPAOH concentration, the C₅⁺ hydrocarbons selectivity is greatly restrained to 5.3% and 8.0%, respectively, while the C₂-C₄ hydrocarbons selectivity increases significantly to about 61%, compared to the supported catalyst (FeMnK/S-1). Unfortunately, they only provide a light olefins selectivity of 31.7% and 38.1% due to the lower olefin/paraffin ratio in C₂-C₄ hydrocarbon (denoted as O/P), which are about 1.1 and 1.6, respectively. A further increase in NaOH concentration to x = 0.20 only slightly affects the selectivity of C₂-C₄ hydrocarbon, but markedly increases the O/P ratio to 5.1, and thus leads to a higher light olefin selectivity about 50.6%. However, the continuous increase of NaOH content from x = 0.20 to x = 0.80 causes the decrease of C₂-C₄ hydrocarbons selectivity and the increase of C₅⁺ hydrocarbons fraction, while maintains O/P at 4–5. In this process, the selectivity of light olefins decreases from 50.6% to 40.5%.

According to the Anderson-Schulz-Flory (ASF) diagram of FeMnK@H-S-1(x) catalysts in Fig. 8b, the chain growth probability (α) increases from 0.40 to 0.54, as the value of x increases from 0.00 to 0.80, demonstrating that larger pores can enhance the chain growth probability. During this process, CO₂ selectivity increases from 16.2% of FeMnK@H-S-1(0.00) to 38.6% of FeMnK@H-S-1(0.20), and then changes little. The suppressed CO₂ selectivity over FeMnK@H-S-1(0.00) and FeMnK@H-S-1(0.10) catalysts might be attributed to the higher H₂/CO ratio on the metal oxide core due to the higher diffusion rate of H₂ than CO in the micropores [40]. The increase in H₂ concentration and the decrease in CO concentration would move the WGS reaction (CO + H₂O ⇌ H₂ + CO₂) equilibrium to the left, thereby inhibiting the production of CO₂ [41,42]. It should be noted that the CO₂ selectivity on the FeMnK@H-S-1(x) (x ≥ 0.20) catalysts is close to that of the supported catalyst (see Table S3). Therefore, we speculated that the more accessible iron oxide in the mesoporous or macroporous catalysts might be responsible for the higher WGS activity.

Regarding the catalytic activity over FeMnK@H-S-1(x) catalysts, a volcanic dependence of the CO conversion on the composition of mixed alkaline solution was observed. With increasing x value from 0.00 to

0.20, the CO conversion increased from 7.5% to 13.9%, which might be attributed to the increase in mass transfer efficiency due to the introduction of mesopores. As the x value further increases from 0.20 to 0.80, the CO conversion decreases slightly from 13.9% to 11.3%. Although the mass transport efficiency increases, the strong metal-support interaction as validated by XPS and H₂-TPR characterizations probably offset an increase in transfer efficiency, resulting in slightly lower CO conversion. As shown in Fig. 8a, the space-time yield (STY) of light olefins also shows a volcanic dependence on the composition of the mixed alkaline solution, reaching a maximum at 26.7 μmol C₂-C₄·g_{Fe}⁻¹·s⁻¹ over FeMnK@H-S-1(0.20) catalyst, consistent with the trend of CO conversion, suggesting that the introduction of appropriate amount of mesopores with certain pore size is beneficial to the catalytic activity and the light olefins selectivity.

The optimized catalytic performance is also compared with that reported in typical literatures. As shown in Table S5, the catalytic activity and light olefins selectivity over the FeMnK@H-S-1(0.20) catalyst are higher than those of similar iron-based encapsulated catalysts (Fe/C@Si-PAPO-K[20] and FeMnK/Al₂O₃@S2[43]) used in CO hydrogenation. The FeMnK@H-S-1(0.20) catalyst in the present work displays a comparable light olefin selectivity to that of Fe/MnK-CNTs[44], Fe-Na-S/CNF[6] and Fe-K/CNTs[9] catalysts, but higher than that of Fe₂₀Mn₁₀/SiO₂[16] catalyst. In addition, even though the FeMnK@H-S-1(0.20) catalyst has lower light olefins selectivity than the reported FeBi/CNT-in and FePb/CNT-in catalysts under similar CO₂ selectivity, the STY of C₂-C₄ olefins of the former catalyst is higher than that of the latter ones [3]. Hence, the FeMnK@H-S-1(0.20) catalyst exhibits superior catalytic performance, being a promising candidate for efficient synthesis of light olefins from syngas.

3.3. Structure-performance investigation

The differences in the catalytic performance of FeMnK@H-S-1(x) catalysts may be stemmed from the diversity of the intrinsic physical and chemical properties of the samples, prompting us to gain deep insights into structure-performance relationship. Therefore, we discussed the structure-performance relationship in detail from the physical structure and chemical properties of the catalyst.

3.3.1. Physical shape-selective effect

As to FeMnK@H-S-1(x) catalysts, the inherent micropores of silicalite-1 endowed FeMnK metal oxide a general shape selectivity effect, which is conducive to limiting the generation of long-chain hydrocarbons. However, the severe transport limitations of the long micropores of zeolite tends to inhibit the diffusion of reactants and products, leading to lower catalytic activity and higher selectivity to

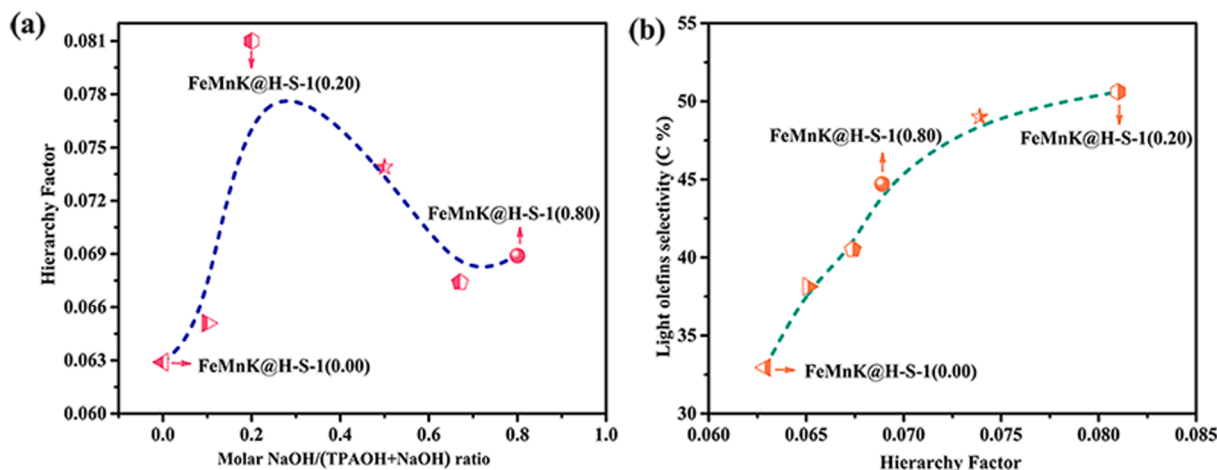


Fig. 9. (a) Hierarchy factor of the FeMnK@H-S-1(*x*) catalysts, defined as the product $(V_{\text{micro}}/V_{\text{total}}) \times (S_{\text{meso}}/S_{\text{BET}})$. (b) The light olefins selectivity versus hierarchy factor.

undesirable low-carbon alkanes (CH_4 and $\text{C}_2^\circ\text{-C}_4^\circ$). The existence of macro-/mesopores ensures high mass-transfer efficiency, makes it easier for reactants to access the active center, and quickly remove the primary light olefins products, thereby improving the catalytic activity and O/P ratio in $\text{C}_2\text{-C}_4$ range. Nevertheless, the introduction of macro-/

mesopores will more or less damage the shape-selective effect of the inherent micropores of the silicalite-1. Therefore, a delicate balance between micropores and macro-/mesopores, that is, between shape-selective ability and transport efficiency should be established to enhance olefin selectivity in CO hydrogenation. To quantify the

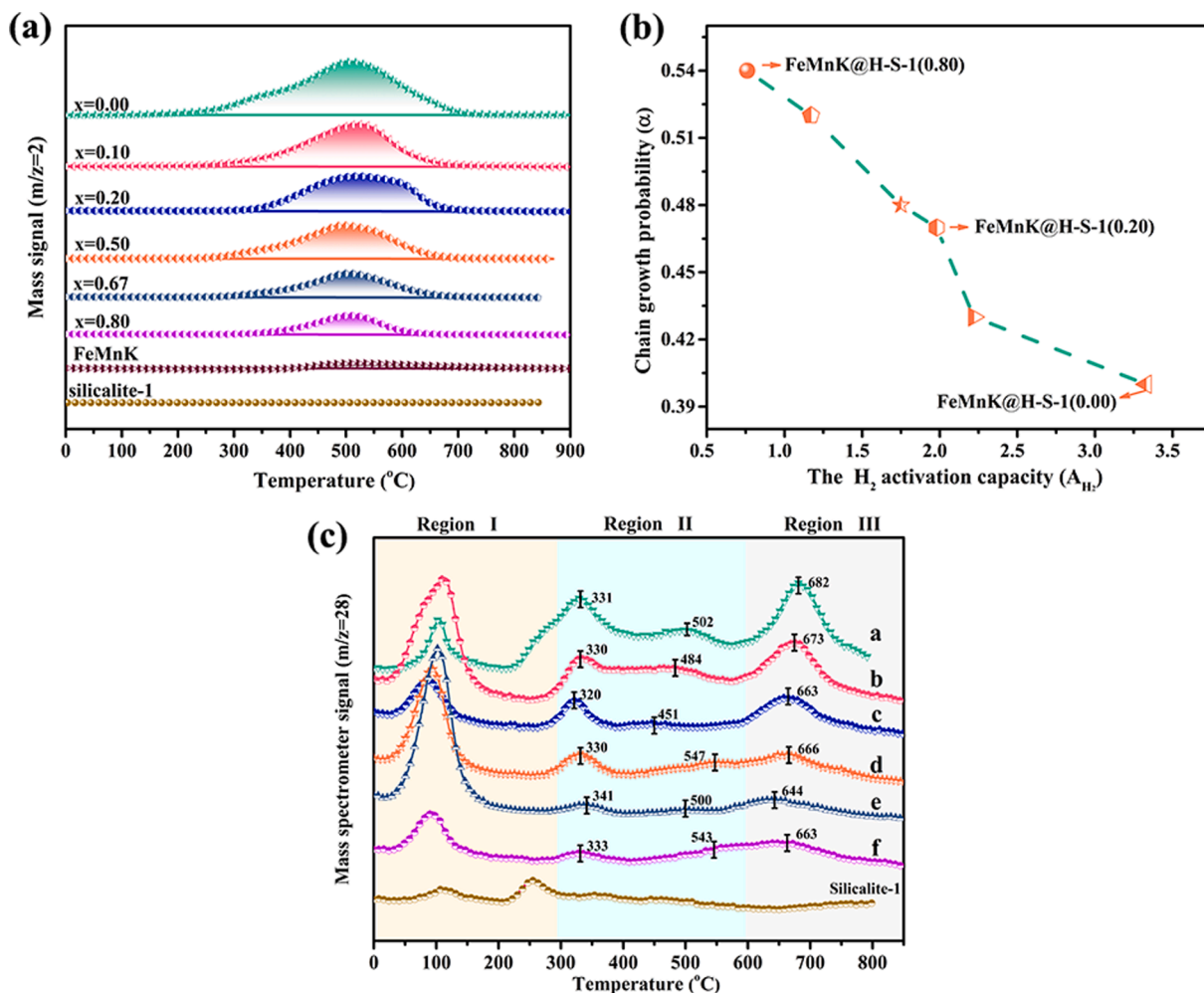


Fig. 10. (a) H_2 -TPD-MS profiles of silicalite-1, reduced FeMnK oxides and FeMnK@H-S-1(*x*) catalysts. (b) Correlations of the chain growth probability (α) with H_2 activation capacity (A_{H_2}). (c) C_2H_4 -TPD-MS profiles of silicalite-1 and reduced FeMnK@H-S-1(*x*) catalysts, a: $x=0.00$; b: $x=0.10$; c: $x=0.20$; d: $x=0.50$; e: $x=0.67$; f: $x=0.80$.

variation of the pore structure of each catalyst with the composition of the alkali solution used, the hierarchical pore factor proposed by Pérez-Ramírez et al. was considered [45]. Hierarchy factor is the result of the relative pore volume of micropores multiplied by the relative specific surface area of mesopores, as shown in the following formula:

$$\text{Hierarchy Factor} = (V_{\text{micro}}/V_{\text{total}}) \times (S_{\text{meso}}/S_{\text{BET}})$$

The parameters V_{micro} , V_{total} , S_{meso} and S_{BET} are derived from the N_2 physical adsorption isotherm. It reflects the balance of micropore volume to the total pore volume and the mesopore surface area to the total surface area of the catalyst. Table S2 summarizes the textural properties of FeMnK@H-S-1(x) catalysts in detail. The hierarchy factor of FeMnK@H-S-1(x) catalysts exhibits a volcano-type relation to the ratio of NaOH/(TPAOH + NaOH), as presented in Fig. 9a. The FeMnK@H-S-1(0.20) catalyst has the highest hierarchy factor, owing to the extensive formation of mesopores coupled with much better preserved micropore volume. However, further increasing NaOH content in the alkaline solution, the hierarchy factor of the obtained FeMnK@H-S-1(x) catalysts ($x \geq 0.50$) decreases, probably due to the reduction of mesopores surface area and micropores volume caused by an increase in the pore diameter of the mesopores or the formation of macropores.

The influence of hierarchy factors on light olefins selectivity could be further excavated to explain the variation of the light olefins selectivity on the FeMnK@H-S-1(x) catalysts. A monotonicity relation between light olefins selectivity and the hierarchy factor of FeMnK@H-S-1(x) catalysts was obtained (Fig. 9b). The light olefins selectivity on FeMnK@H-S-1(0.20) catalyst with higher mesopore surface area is better than that of FeMnK@H-S-1(0.00) and FeMnK@H-S-1(0.10) catalysts with higher micropore volume. It suggested that the benefits of improved transport efficiency in the former catalyst owing to the elevated mesoporous area are more influential than the expense of decreased microporosity, and the net effect leads to superior catalytic performance. Thereby the light olefins selectivity was robust through the enhanced transport efficiency and suppressed re-hydrogenation of light olefins. But for FeMnK@H-S-1(x) ($x \geq 0.50$) catalysts, the light olefins selectivity are reduced, caused by the increasingly damaged shape-selective effect, resulting in higher C_5^+ hydrocarbon selectivity. Interestingly, the space-time yield (Fig. S3) of light olefin also shows a monotonically increasing dependence on the hierarchy factor. This experiment shows that, in light olefins synthesis over the metal@hierarchical-zeolite system, the porosities should be optimally orchestrated to accomplish complementary missions in the catalyst: micropores provide a shape-selective effect, while the auxiliary mesopore networks improve the accessibility of active metals and molecular transport. More importantly, it reveals that the maximization of the hierarchy factor seems to be essential to attain higher light olefins selectivity over FeMnK@H-S-1(x) catalysts in FTO process.

3.3.2. Chemisorption behaviors of FeMnK@H-S-1(x) catalysts

To further gain insights into the different catalytic performance of FeMnK@H-S-1(x) catalysts, the chemisorption behaviors of H_2 and C_2H_4 were also investigated. Fig. 10a presents the corresponding hydrogen-temperature programmed desorption-mass spectroscopy (H_2 -TPD-MS) curves. It can be seen that all FeMnK@H-S-1(x) catalysts present a H_2 desorption peak and the desorption amount decreases as the pore size increases. In addition, naked FeMnK exhibits a weaker hydrogen desorption peak than the aforementioned catalysts, and no obvious hydrogen desorption peak is found on silicalite-1 support, which suggests that the combination of the two, rather than any part, takes the main responsibility for the enhanced and regular hydrogen desorption capacity, drawing forth the H_2 spillover phenomenon. It has also been found that H_2 spillover phenomenon could occur on supported catalysts, being attributed to the overflow of H_2 from strong chemisorption sites to the catalyst support [46,47]. Besides, the degree of hydrogen spillover was greatly influenced by the surface area of the support [48]. As pore size increases, the surface area of the investigated catalysts reduce,

causing the decrease of H_2 adsorption/desorption amount. In our present work, some sodium ions could not be completely washed out in the preparing process of the catalysts, as shown in Table S1. According to the previous studies, Na is supposed to be one of the typical electronic promoters, which can also inhibit the adsorption of H_2 [49]. Thus, the effect of residual Na on H_2 adsorption cannot be ruled out. Based on the above discussion, it could be inferred that the H_2 spillover effect and residual Na of the catalysts together determine the reduction in hydrogen desorption capacity.

The desorption amount of hydrogen seems to be well in coincidence with the hydrocarbon distribution, and a monotonic relationship between hydrogen activation capacity and hydrocarbon distribution is established in Fig. 10b, where the index of H_2 activation capacity (denoted as A_{H_2}) is defined as desorption peak area of H_2 from Fig. 10a. As shown in Fig. 10b, with the decrease of A_{H_2} , the chain growth probability (α) increases from 0.40 to 0.54. Specifically, a higher H_2 activation capacity (A_{H_2}) provides more opportunities for the hydrogenation reaction, facilitating chain termination reaction and then promoting the formation of low-carbon alkanes, especially methane. On the contrary, a lower H_2 activation capacity (A_{H_2}) inhibits the chain termination reaction caused by alkene hydrogenation, and promotes carbon chain growth to produce C_5^+ hydrocarbons. Consequently, the moderate H_2 activating capacity is conducive to the production of light olefins.

The interactions of ethylene with the reduced FeMnK@H-S-1(x) catalysts were studied by means of ethylene-temperature programmed desorption-mass spectroscopy (C_2H_4 -TPD-MS). As presented in Fig. 10c, the C_2H_4 -TPD desorption profiles of FeMnK@H-S-1(x) catalysts are mainly distributed in three regions of 0–300 °C (I), 300–600 °C (II) and 600–800 °C (III), attributing to the weak, moderate and strong adsorption of ethylene. It is worth noting that the obvious ethylene signal is observed in region I over silicalite-1 support. Therefore, desorption of ethylene in region I might be due to the weak adsorption on silicalite-1 zeolite. The desorption peaks in region II and III might be attributed to the C_2H_4 adsorbed on iron species, as confirmed by C_2H_4 -TPD profiles of reduced FeMnK oxides (see Fig. S4). It can be seen that the desorption curve of FeMnK@H-S-1(0.20) catalyst shows the lowest desorption temperature centered at 320 and 451 °C, revealing a weaker ethylene and iron species interaction for the catalyst. The promoted ethylene desorption over FeMnK@H-S-1(0.20) catalyst indicates that most of C_2H_4 could be easier to desorb once it was generated due to the lower ethylene desorption energy, thereby preventing the continued carbon chain propagation or hydrogenation reaction of ethylene and improving the selectivity of light olefins. These results are in good consistence with the above catalytic performance.

3.3.3. Exclusion of the sodium influence

FeMnK@H-S-1(x) catalysts were obtained by treating FeMnK/S-1 with different alkali solutions ($x = \text{NaOH}/(\text{TPAOH} + \text{NaOH})$). According to the XRF characterization (Table S1), as the x value increased from 0.00 to 0.80, the residual Na content varied more or less. Previous literature reports that alkali metal is a good electronic promoter, which could enhance the olefins selectivity for the iron-based FT synthesis catalysts [8,50,51]. In order to eliminate the influence of Na promoter, we synthesized Na modified FeMnK@H-S-1(0.00) catalyst with Na content of 1.17 wt% (see Table S6 and S1) similar to the FeMnK@H-S-1(0.20, 0.50, 0.67) catalysts. Compared with the original FeMnK@H-S-1(0.00) catalyst, the Na modified catalyst exhibits slightly enhanced light olefins selectivity (33.9%), and its value is still lower than that of the FeMnK@H-S-1(0.20, 0.50, 0.67) catalysts (Table S7). The above experimental results demonstrated that the promotion effect of Na is not the main reason for the disparate catalytic performance of FeMnK@H-S-1(x) catalysts.

3.3.4. C_2H_4 pulse experiment

An important factor in improving the light olefins selectivity is

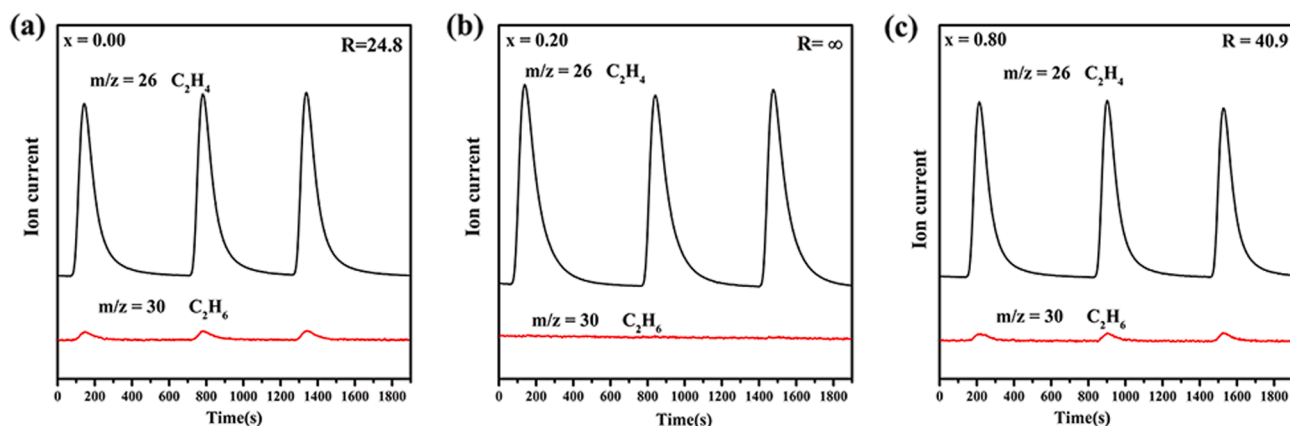


Fig. 11. Transient response curves of used FeMnK@H-S-1(x) catalysts detected by MS (a) 0.00, (b) 0.20, (c) 0.80 by pulse of 0.8 mL C_2H_4 into stable H_2 flow at 280 °C. $R = \text{Peak area of } C_2H_4 / \text{Peak area of } C_2H_6$.

reducing olefins secondary reaction through their re-adsorption in the diffusion downstream. From the diversity of the light olefins selectivity on the FeMnK@H-S-1(x) catalysts, we speculate that the secondary reaction probability of the samples might be different. To confirm this speculation, we simulated olefins secondary reaction by C_2H_4 hydrogenation, where the pulse experiment of C_2H_4 was performed to test the re-adsorption and hydrogenation odds on the used FeMnK@H-S-1(0.00), FeMnK@H-S-1(0.20) and FeMnK@H-S-1(0.80) catalysts. First, the used catalyst was treated in argon flow to remove surface residues, and then switched to H_2 atmosphere, after which 0.8 mL of C_2H_4 was pulsed and the outlet gas was analyzed by MS analyzer. A part of ethylene was hydrogenated into ethane in this process, and the degree of hydrogenation can be judged from the ratio of ethylene to ethane in exit gas, denoted as R. As shown in Fig. 11a, the FeMnK@H-S-1(0.00) catalyst has an obvious ethane response peak, with a lower R value of 24.8, indicating that it has strong second hydrogenation ability. However, the formation of ethane on the FeMnK@H-S-1(0.20) catalyst is almost completely inhibited (see Fig. 11b), which validates that the olefin re-adsorption or hydrogenation on this catalyst is strongly suppressed. Furthermore, the FeMnK@H-S-1(0.80) catalyst also exhibits apparent ethane response peak, which might be due to the higher availability of active metals for ethylene molecules adsorption on the catalyst with open macropores. Combined with the physical and chemical characterization as aforementioned, the high light olefins selectivity of FeMnK@H-S-1(0.20) catalyst may be caused by the followed reasons: (1) well-orchestrated hierarchical zeolite shell; (2) moderate hydrogen activation capacity; (3) lower olefins desorption energy; (4) reduced olefins re-adsorption or secondary reaction probability.

4. Conclusion

In summary, we have fabricated a series of hierarchical zeolite encapsulated FeMnK catalysts to enhance light olefins selectivity in Fischer-Tropsch synthesis. Our experiments confirmed that the texture of poly-type hierarchical pore encapsulated catalysts was precisely regulated by adjusting mixed alkaline solution during the catalyst preparation. In the dynamic change stages involved micropore, mesopore and macropore, the olefins selectivity showed a volcanic curve with the increase of pore size and reached a maximum of 50.6% on FeMnK@H-S-1(0.20) catalyst. An interesting monotonicity relationship between light olefins selectivity and the hierarchy factor of FeMnK@H-S-1(x) catalysts was firstly declared, indicating that maximization of the hierarchy factor was essential to attain higher light olefins selectivity, due to a good balance between the shape selectivity of micropores and the mass transfer efficiency of mesopores. In addition, the moderate H_2 activation capacity and relatively lower olefins desorption energy could

also be responsible for the high light olefin selectivity of the FeMnK@H-S-1(0.20) catalyst. Moreover, the olefins secondary reaction via re-adsorption was obviously suppressed. The present work reflects the significance of the zeolite shell texture in modulating light olefins selectivity and will be beneficial to construct effective CO hydrogenation catalysts.

CRediT authorship contribution statement

Can Zhu: Conceptualization, Investigation, Methodology, Software, Writing – original draft, Visualization. **Chao Huang:** Software, Resources, Writing - review & editing, Methodology. **Mingwei Zhang:** Project administration, Conceptualization, Writing - review & editing, Validation, Supervision. **Yizhuo Han:** Conceptualization, Supervision. **Kegong Fang:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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